

Assignment 1: DDD contracts

Due Tuesday, October 6, 23:59 on Brightspace

This assignment are submitted in teams of 2, with one submission per team. Identify all authors on the cover page of the submission by giving their names and student numbers.

Any submission of which you are not the author is a violation of academic integrity. This includes solutions provided by any third party, including those generated by artificial intelligence tools.

Background

An airport with hundreds of employees is planning to replace its current employee management process, which relies on physical forms and spreadsheets, with a digital system called the Airport Employee Management System (AEMS). AEMS will integrate with the airport's existing payroll system and a new AI-based intelligent employee scheduling system that the airport recently purchased.

The intelligent employee scheduling system dynamically creates weekly work schedules based on several factors, including employee preferences and expected airport traffic. However, the scheduling system does not currently account for regulatory constraints, such as the maximum number of hours an employee can work in a shift and the minimum number of hours required between consecutive shifts. AEMS is intended to address this limitation by ensuring that generated schedules comply with applicable regulatory constraints. As part of the process, AEMS also needs to record employee scheduling preferences. The airport plans to release the first version of AEMS as soon as possible and continue to expand and improve the system over time.

Anyone who uses AEMS must log in to the system. AEMS records various types of information about each employee, including their first name, last name, date of birth, address, job title, Social Insurance Number (SIN), and banking information. An employee's address consists of a street address, city, postal code, and an optional unit number. Banking information includes an account number, branch number, and transit number.

The airport employs people in different roles, including Check-in Agent, Gate Agent, Baggage Handler, Ramp Agent, Security Screener, Maintenance Technician, Shuttle Driver, Customer Service Agent, and Cleaner. The airport is organized into several departments: Airport Management, Passenger Services, Ground Operations, Security, Maintenance, and Ground Transportation. Each department is managed by a Department Manager.

For the first version of AEMS, the airport plans to deploy the system only to customer-facing departments. In future versions, the airport intends to expand AEMS to support all departments across the airport.

Questions

Question 1 (8 marks)

Identify the core subdomain of the business domain of AEMS. Identify four additional subdomains and label them as generic or supporting. Justify your labelling in one or two sentences.

Question 2 (40 marks)

Identify the application command(s) arising from use cases *Register Department Employee* and *Select Work Shifts* of AEMS, indicate the step where each command occurs, and provide DDD contracts for these.

You are only required to provide contracts for domain-significant application commands (those that result in changes to the domain).

State your assumptions, if any.

Consider the following use cases, developed as part of the requirements for AEMS.

Use case *Register Department Employee* is performed when registering a new employee.

Use case: Register Department Employee	
Primary actor	Airport Manager
Precondition	Airport Manager is authenticated
Steps	1. The Airport Manager selects to register an Employee. 2. The AEMS asks for the employee registration information. 3. The Airport Manager submits the employee registration information. 4. The AEMS registers the employee and displays the id number of the new employee. 5. The AEMS presents the list of departments. 6. The Airport Manager selects the department in which the new employee is assigned. 7. The AEMS assigns the employee to the selected department and notifies the department's manager.
Postcondition	A new employee was registered in a department and the department's manager was notified

Use case *Select Work Shifts* is performed when an employee wants to indicate their preferences for upcoming shifts. The list of shifts is defined by the department manager. Each shift is specified by its date, shift start time, and shift end time.

Use case: Select Work Shifts	
Primary actor	Employee
Precondition	Employee is authenticated
Steps	1. The Employee asks to see the list of upcoming shifts in their department.

	2. The AEMS displays the list of shifts. 3. The Employee selects the shifts for which they are available and submits. 4. The AEMS records the employee's selection.
Postcondition	The preferred shifts of the employee were recorded

The intelligent scheduling system subsequently creates the weekly work schedule of the department, taking the recorded preferences into account.

Question 3 (2 marks)

Which of the previous two use cases would you place in the same bounded context as the core subdomain of the system? Justify your answer.

Marking scheme

Question	Item	Marks
Question 1 (8 marks)	Identifying the core subdomain	4
	Each additional subdomain identified (four in total)	1 each
Question 2 (40 marks)	Commands and contracts of <i>Register Department Employee</i>	30
	Commands and contracts of <i>Select Work Shifts</i>	10
Question 3 (2 marks)	Identifying and justifying the use case	2
Total		50